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Prediction of Diagnosis Related Groups from Patient Diagnoses, Procedures, Age, and Sex

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This work was developed as part of a machine learning course project using anonymized Diagnosis Related Group data from Hospital El Pino.

ABSTRACT Diagnosis Related Groups (DRG), referred to as Grupos Relacionados por Diagnostico (GRD) in the Chilean hospital context, classify hospital discharges according to clinical similarity, severity, and expected resource consumption. This project addresses GRD prediction from structured patient information from Hospital El Pino, including diagnosis codes, procedure codes, age, sex, and the final assigned GRD. The task is formulated as a high-cardinality and highly imbalanced multiclass classification problem. Exploratory data analysis identified 14,561 records, 526 original GRD labels, 297 classes with nine or fewer observations, and only 32 classes with more than 100 observations. These properties make accuracy alone insufficient and motivate the use of macro-F1, weighted-F1, and top-3 accuracy. The proposed workflow combines code extraction, target filtering, stratified splitting, feature-combination search, and the comparison of two machine learning approaches: a neural network with learned embeddings and a Random Forest baseline. After filtering classes with at least 30 cases, 11,480 records and 113 GRD classes were retained. The final experiments show that the neural network with embeddings outperformed the Random Forest baseline, reaching 0.8528 accuracy, 0.7964 macro-F1, 0.8488 weighted-F1, and 0.9617 top-3 accuracy. The best Random Forest configuration reached 0.7108 accuracy, 0.6199 macro-F1, 0.7020 weighted-F1, and 0.9264 top-3 accuracy. These results suggest that learned embeddings are more suitable for representing sparse high-cardinality medical codes than rigid encodings used by tree-based models.

INDEX TERMS Clinical coding, class imbalance, Diagnosis Related Groups, DRG, embeddings, GRD, high-cardinality categorical variables, machine learning, multiclass classification, Random Forest.

I. INTRODUCTION

DIAGNOSIS Related Groups (DRGs) are classification systems that group hospital cases according to clinical conditions, procedures, severity, and expected resource consumption. In Chile, the GRD mechanism has become relevant for hospital management and public financing, particularly in reimbursement and activity-monitoring processes associated with FONASA [2]. Correct GRD assignment is therefore clinically and administratively relevant: it affects hospital statistics, planning, resource allocation, and payment mechanisms.

Manual GRD assignment is generally retrospective and depends on the completeness of clinical documentation and coding quality. Machine learning can support this process by estimating GRD labels from structured or unstructured patient information before manual coding is finalized. Prior work has shown that automated DRG prediction can support early operational decision-making and resource planning [1]. This project focuses on the structured-data setting, where diagnoses and procedures are available as categorical clinical codes.

The problem is challenging because both the predictors and the target have high cardinality. Diagnosis and procedure codes are sparse categorical variables, and the GRD target distribution is strongly imbalanced. This work compares two model families: a neural network using embeddings for clinical code representation and a Random Forest baseline using grouped and encoded medical codes. The study also performs a feature-combination search to determine how many secondary diagnosis and procedure fields should be included as inputs.

II. RELATED WORK

Prior research has explored machine learning for DRG prediction as a way to reduce the delay and inefficiency of manual hospital coding. Liu et al. used deep learning over clinical notes to predict DRGs and hospital costs before discharge, showing that DRG prediction can support proactive resource planning [1]. In contrast, the present project does not use free-text clinical notes; it focuses on structured diagnosis and procedure codes, which makes categorical representation a central methodological issue.

In the Chilean context, Pena-Torres [2] describes the role of GRD-based payment and public hospital management. That work does not propose a predictive model, but it provides institutional motivation for automating and improving GRD-related processes. More recent neural approaches, such as DRG-LLaMA [3], use large language models for DRG prediction from clinical text. Compared with those approaches, this project evaluates whether learned dense representations are useful even when the available inputs are structured clinical codes rather than full clinical narratives.

Direct numerical comparison with previous studies is limited because related work uses different hospitals, coding systems, data modalities, and target definitions. Nevertheless, the methodological trend is consistent: learned representations are useful when clinical information is high-dimensional, sparse, or semantically structured.

III. STUDY OBJECTIVE

The general objective is to develop and evaluate a machine learning model capable of predicting the GRD of a hospital case from patient-level structured information. The specific objectives are:

1) To characterize the dataset through exploratory data analysis, including completeness, correctness, descriptive statistics, class cardinality, and class imbalance.
2) To design a preprocessing pipeline for high-cardinality diagnosis and procedure codes.
3) To experimentally justify the selected diagnosis and procedure inputs through feature-combination search.
4) To compare at least two machine learning techniques suitable for multiclass GRD prediction.
5) To evaluate performance using metrics appropriate for imbalanced multiclass classification, prioritizing macro-F1 over accuracy alone.

IV. DATASET DESCRIPTION

The dataset corresponds to anonymized GRD records from Hospital El Pino. Each row represents a patient case and includes demographic variables, clinical diagnosis fields, clinical procedure fields, and the final assigned GRD. Diagnosis codes follow the CIE-10 structure, while procedure codes follow the CIE-9 structure.

The original dataset contained 14,561 records and 68 columns. The target had 526 unique GRD classes. Table 1 shows the five most frequent classes observed before target filtering. The target distribution is long-tailed: 297 classes had nine or fewer observations, while only 32 classes had more than 100 observations. This makes the original 526-class problem unsuitable for robust stratified training without target filtering.

TABLE 1. Most Frequent GRD Classes in the Original Dataset

GRD class	Count
146101 – PH Cesarean section	813
146121 – PH vaginal delivery with procedure	639
146131 – PH vaginal delivery	538
158171 – MH neonate, birth weight >2499 g	389
134161 – MH antepartum disorders	325

V. METHODOLOGY

The methodology follows a supervised machine learning workflow. First, the dataset is inspected to identify target cardinality, duplicate records, clinical field structure, demographic distributions, and sparse secondary-code positions. Second, code-description fields are normalized by extracting only the code component. Third, rare target classes are filtered using a minimum-frequency threshold. Fourth, training and test partitions are created through stratified splitting to preserve the retained class distribution. Fifth, feature-combination search evaluates different numbers of secondary diagnosis and procedure fields. Finally, the best neural network and Random Forest candidates are compared using the same test set and metrics.

A. DATA QUALITY ASSESSMENT

The initial audit found 14,561 records, 68 columns, and 144 duplicated rows. These duplicated rows were not automatically removed because the dataset represents hospital cases and identical coded profiles may occur in real clinical-administrative records. Completeness was assessed by checking explicit null values and the presence of structural absence markers. In secondary diagnosis and procedure fields, the symbol "-" indicates that the patient did not have an additional code in that position; therefore, it was treated as structural absence rather than as an erroneous missing value.

Correctness was evaluated by checking the expected code-description structure of diagnosis, procedure, and GRD fields before extracting the code component. Outlier analysis focused on age, the only numerical predictor retained for modeling. The observed age range was 0 to 121 years, covering neonatal and elderly cases. No additional numerical feature with comparable outlier risk was used as a predictor.

B. PREPROCESSING STRATEGY

The preprocessing strategy was tailored to each model family. For the neural network, diagnosis and procedure codes were mapped to integer indices and represented through embedding layers. This approach avoids the dimensional explosion of one-hot encoding and allows the model to learn dense representations of high-cardinality medical codes. The final selected neural model used embedding dimensions of 32

for both diagnosis and procedure codes. A fixed number of diagnosis and procedure slots was selected through feature-combination search, while empty secondary positions were encoded as structural absence.

For the Random Forest baseline, categorical medical codes were reduced to broader code prefixes and then encoded into numerical representations compatible with tree-based learning. Diagnosis codes were grouped by their first three characters, while procedure codes were grouped by a two-character prefix after removing punctuation. This grouping reduces cardinality but loses some clinical granularity compared with the embedding-based neural representation.

C. TARGET FILTERING AND CLASS IMBALANCE

The original target contained 526 GRD classes, including many labels with extremely low support. To allow stratified splitting and meaningful class-level evaluation, the final experiments retained only classes with at least 30 observations. This threshold produced a final modeling dataset of 11,480 records and 113 GRD classes, with a minimum class support of 30 and a maximum support of 813.

Class imbalance was addressed primarily through target filtering, stratified splitting, class-weighted neural variants, and evaluation with macro-F1. SMOTENC was considered during exploratory analysis as an additional strategy for mixed categorical and numerical data, but it was not selected as the final strategy because synthetic clinical records may introduce unrealistic diagnosis-procedure combinations if not clinically validated.

D. FEATURE-COMBINATION SEARCH

A central experimental component is the search over diagnosis and procedure inputs. Instead of assuming that all available secondary fields are useful, the notebook evaluates different numbers of secondary diagnosis fields and secondary procedure fields. Let d be the number of secondary diagnosis fields and p the number of secondary procedure fields. The search evaluates:

$$(d, p) \in D \times P, \tag{1}$$

where D and P are predefined candidate sets. Each pair is trained and evaluated using the same split and metrics. The best neural-network feature configuration was selected by macro-F1 and used 16 secondary diagnosis fields and 10 secondary procedure fields.

VI. MACHINE LEARNING MODELS

A. NEURAL NETWORK WITH EMBEDDINGS

The neural network receives diagnosis slots, procedure slots, age, and sex as inputs. Each diagnosis and procedure slot is encoded as an integer and passed through an embedding layer. The embedded vectors are flattened and concatenated with the demographic variables. The resulting representation is passed to a dense classifier and a softmax output layer with one unit per retained GRD class.

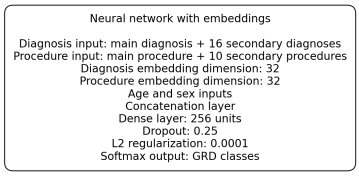

FIGURE 1. Architecture of the selected neural network with embeddings. Diagnosis and procedure slots are represented through embedding layers and concatenated with age and sex before dense classification.

The selected model used 16 secondary diagnosis fields, 10 secondary procedure fields, diagnosis and procedure embeddings of dimension 32, one dense layer with 256 units, ReLU activation, dropout of 0.25, and L2 regularization of 10^{-4}. It was trained with sparse categorical cross-entropy, Adam optimization, early stopping, and learning-rate reduction on validation loss.

B. RANDOM FOREST BASELINE

The Random Forest baseline receives grouped and label-encoded diagnosis and procedure categories, together with age and sex. The model builds an ensemble of decision trees trained over bootstrap samples and random subsets of features. It is appropriate as a non-neural comparison because it can model nonlinear interactions after encoding. However, Random Forest does not learn distributed representations of clinical codes and may be less expressive for high-cardinality sparse categorical variables than embeddings.

VII. EVALUATION METRICS

The evaluation uses multiple metrics due to class imbalance. Accuracy is reported as a global indicator, but it can overestimate performance when frequent classes dominate. Macro-F1 is the primary selection metric because it gives equal weight to every retained GRD class. Weighted-F1 is reported to reflect performance weighted by support. Top-3 accuracy is included because a decision-support system may still be useful if the correct GRD appears among the three most likely alternatives.

For class c, precision, recall, and F1-score are:

$$Precision_c = \frac{TP_c}{TP_c + FP_c}, \tag{2}$$

$$Recall_c = \frac{TP_c}{TP_c + FN_c}, \tag{3}$$

$$F1_c = 2 \cdot \frac{Precision_c \cdot Recall_c}{Precision_c + Recall_c}. \tag{4}$$

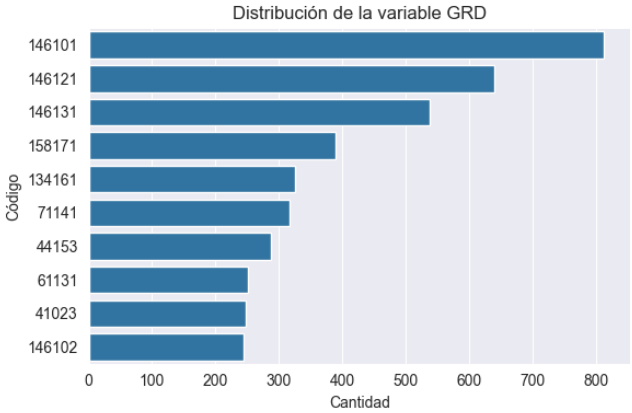

FIGURE 2. Distribution of the most frequent GRD classes.

Macro-F1 is computed as:

$$MacroF1 = \frac{1}{C}\sum_{c=1}^{C} F1_c, \tag{5}$$

where C is the number of retained GRD classes.

VIII. EXPLORATORY DATA ANALYSIS

The exploratory analysis identified three main structural properties. First, the target was high-cardinality, with 526 original GRD classes. Second, the target was strongly long-tailed, as shown in Fig. 2, Fig. 3, and Fig. 4. Third, secondary diagnosis and procedure fields became increasingly sparse as the position increased.

Table 2 summarizes representative diagnosis fields. The first diagnosis field had 1,491 unique values, while later secondary diagnosis fields increasingly used the structural absence marker "-". This supports the decision to evaluate different numbers of secondary fields rather than including all columns automatically.

TABLE 2. Descriptive Statistics for Representative Diagnosis Columns

Variable	Count	Unique	Top	Freq.
Diag 01 Principal	14561	1491	O70.0	779
Diag 02 Secondary	14561	1717	-	1209
Diag 03 Secondary	14561	1498	-	2687
Diag 04 Secondary	14561	1304	-	4127
Diag 05 Secondary	14561	1151	-	5371

IX. EXPERIMENTS AND RESULTS

The experiments compare neural-network candidates and Random Forest candidates under the same evaluation criteria. The neural-network feature search used an embedding architecture with two dense layers as a controlled base model for selecting clinical input slots. Final neural candidates then evaluated different architectures, embedding sizes, class-weight strategies, focal loss, dropout, and L2 regularization. Random Forest experiments evaluated grouped-code inputs and ten hyperparameter configurations.

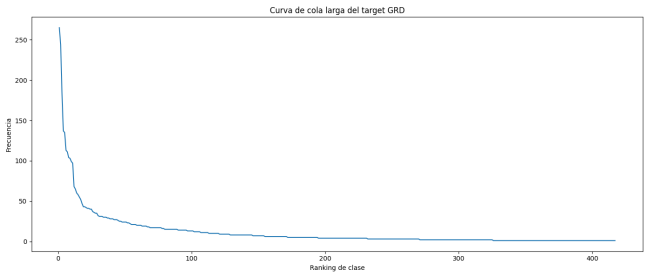

FIGURE 3. Long-tail distribution of GRD class frequencies.

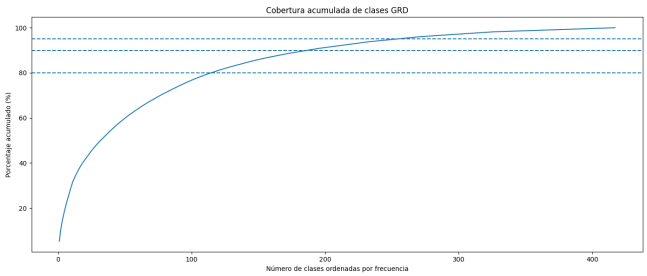

FIGURE 4. Cumulative concentration of the most frequent GRD classes.

A. FEATURE INPUT SELECTION

The neural feature-combination search selected 16 secondary diagnosis fields and 10 secondary procedure fields. This result suggests that a moderate amount of secondary clinical information improves prediction, while including later sparse positions may add noise. For Random Forest, the best final configuration used six secondary diagnosis fields and no secondary procedure fields, indicating that the tree-based representation benefited from a more compact grouped-code input.

B. FINAL MODEL COMPARISON

The final model comparison is summarized in Table 3. The neural network outperformed the Random Forest baseline in all reported metrics. The improvement is especially relevant in macro-F1, which is the primary metric for this imbalanced multiclass task.

C. CANDIDATE MODEL SELECTION

Table 4 reports the ten best neural-network candidates from the final experiments. The best model was the L2-regularized embedding network. This indicates that, after selecting the clinical input slots, controlling overfitting was more beneficial than increasing model depth.

Table 5 reports the ten Random Forest candidates. The best Random Forest used 500 estimators, maximum depth 15, one minimum sample per leaf, square-root feature sampling, and balanced subsampling.

D. TRAINING PROCESS VISUALIZATION

The training curves of the selected neural network are shown in Fig. 5 and Fig. 6. The validation loss decreases during the first epochs and then stabilizes, while validation accuracy

TABLE 3. Final Model Comparison

Model	Acc.	Macro-F1	Weighted-F1	Top-3
Neural network	0.8528	0.7964	0.8488	0.9617
Random Forest	0.7108	0.6199	0.7020	0.9264

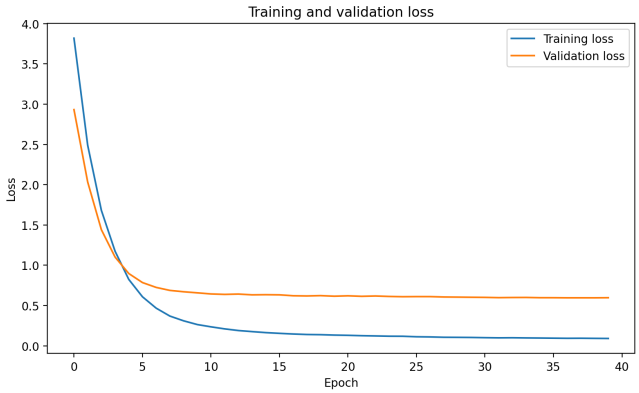

FIGURE 5. Training and validation loss of the selected neural network.

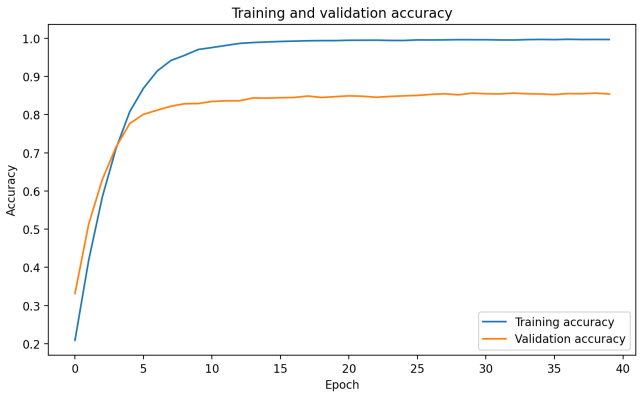

FIGURE 6. Training and validation accuracy of the selected neural network.

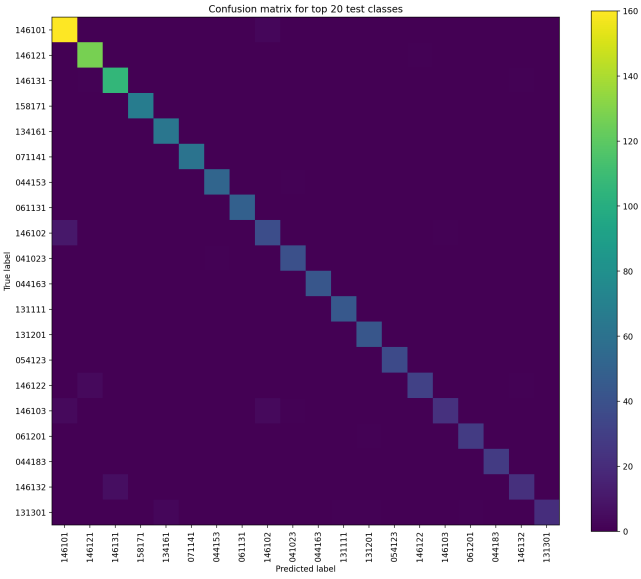

FIGURE 7. Confusion matrix for the 20 most frequent GRD classes in the test set using the selected neural network.

remains below training accuracy. This behavior indicates that the model learns the dominant structure of the data, although a moderate generalization gap remains due to the complexity and imbalance of the task.

E. PERFORMANCE ANALYSIS

A complete confusion matrix for 113 retained classes is difficult to interpret visually. Therefore, Fig. 7 presents the confusion matrix for the 20 most frequent GRD classes in the test set. This view connects the numerical metrics with concrete classification errors among the classes with the largest test support.

X. DISCUSSION

The results support the hypothesis that neural networks with embeddings are better suited to this GRD prediction task than Random Forest models using grouped and encoded categorical variables. Diagnosis and procedure codes are high-cardinality categorical variables; rigid encodings either lose granularity or impose artificial numerical relationships. Embeddings provide compact dense vectors and allow the model to learn statistical similarities between codes during training.

The feature-combination search is also important. Later secondary diagnosis and procedure positions are increasingly sparse, so including every column can add noise and increase model complexity. The best neural configuration used a substantial but not maximal number of secondary clinical slots, while the best Random Forest benefited from a much more compact grouped-code representation. This confirms that the optimal input structure depends on the representation and model family.

The use of macro-F1 as the main selection metric is justified by the long-tailed GRD distribution. Accuracy alone would favor frequent classes and obscure minority-class errors. Top-3 accuracy complements macro-F1 by measuring whether the correct GRD appears among the most likely alternatives, which is useful in a decision-support context.

XI. LIMITATIONS

This study has several limitations. First, the dataset is highly imbalanced, requiring rare-label filtering and reducing the scope of the final prediction task from 526 to 113 classes. Second, the data comes from a single hospital, so the learned patterns may reflect local coding practices and may not generalize directly to other institutions. Third, the model uses structured codes only; it does not incorporate free-text clinical notes, laboratory results, medications, discharge summaries, or temporal information. Fourth, synthetic balancing methods such as SMOTENC were not selected as final components because synthetic clinical combinations require clinical validation. Finally, the evaluation was performed on

TABLE 4. Top Neural-Network Candidates Ranked by Macro-F1

Rank	Model	Dense layers	Emb.	Loss	Reg.	Acc.	Macro-F1	Weighted-F1	Top-3
1	NN_l2	(256)	32/32	CE	L2	0.8528	0.7964	0.8488	0.9617
2	NN_512_cw_smooth	(512)	24/20	CE	smooth CW	0.8389	0.7810	0.8362	0.9625
3	NN_focal	(256)	24/20	focal	none	0.8415	0.7804	0.8386	0.9639
4	NN_focal_embeddings	(256)	32/32	focal	none	0.8415	0.7804	0.8386	0.9639
5	NN_256	(256)	24/20	CE	none	0.8415	0.7791	0.8369	0.9608
6	NN_embeddings_32_32	(256)	32/32	CE	none	0.8415	0.7791	0.8369	0.9608
7	NN_512	(512)	24/20	CE	none	0.8380	0.7786	0.8347	0.9608
8	NN_report_512_cw	(512)	24/20	CE	balanced CW	0.8310	0.7783	0.8306	0.9639
9	NN_256_cw_smooth	(256)	24/20	CE	smooth CW	0.8332	0.7690	0.8304	0.9621
10	NN_embeddings_32_32_cw	(256)	32/32	CE	smooth CW	0.8332	0.7690	0.8304	0.9621

TABLE 5. Top Random Forest Candidates Ranked by Macro-F1

Rank	Model	Key hyperparameters	Diag. sec.	Proc. sec.	Acc.	Macro-F1	Weighted-F1	Top-3	Prec.	Recall
1	RF_depth15_leaf1	500 trees, depth 15, leaf 1	6	0	0.7108	0.6199	0.7020	0.9264	0.6450	0.6226
2	RF_log2	400 trees, log2 features	6	0	0.7043	0.6167	0.6975	0.9238	0.6325	0.6243
3	RF_depth30	400 trees, depth 30	6	0	0.7038	0.6163	0.6970	0.9238	0.6324	0.6239
4	RF_300_balanced	300 trees, balanced	6	0	0.6995	0.6139	0.6940	0.9268	0.6299	0.6219
5	RF_500_balanced	500 trees, balanced	6	0	0.6990	0.6112	0.6930	0.9260	0.6290	0.6184
6	RF_depth25_leaf1	500 trees, depth 25, leaf 1	6	0	0.7108	0.6110	0.6953	0.9321	0.6394	0.6055
7	RF_leaf1	400 trees, leaf 1	6	0	0.7051	0.6081	0.6889	0.9294	0.6393	0.6018
8	RF_depth20	400 trees, depth 20	6	0	0.6964	0.6041	0.6900	0.9260	0.6188	0.6119
9	RF_leaf3	400 trees, leaf 3	6	0	0.6816	0.6001	0.6794	0.9225	0.6170	0.6133
10	RF_no_weight	400 trees, no class weight	6	0	0.6969	0.5692	0.6743	0.9172	0.6219	0.5607

an internal stratified split, so external validation is required before operational use.

XII. FUTURE WORK

Future work should prioritize external validation using data from other hospitals. The model could also incorporate hierarchical information from CIE-10 and CIE-9 taxonomies, either as auxiliary features or as structured embeddings. In addition, attention-based or permutation-invariant architectures could be explored because secondary diagnoses and procedures are not naturally ordered. Finally, explainability methods should be incorporated to identify which diagnosis and procedure codes most influence each predicted GRD.

XIII. CONCLUSION

This project addresses GRD prediction as a high-cardinality, imbalanced multiclass classification problem using structured patient information from Hospital El Pino. The exploratory analysis identified 526 original GRD labels, a pronounced long tail, and many low-support classes. These findings justified target filtering, stratified splitting, feature-combination search, and the use of macro-F1 as the main evaluation metric.

The final results indicate that the neural network with embeddings outperformed the Random Forest baseline, reaching 0.7964 macro-F1 compared with 0.6199 for Random Forest. The best neural configuration used 16 secondary diagnosis fields and 10 secondary procedure fields, embedding dimensions of 32 for both diagnosis and procedure codes, one dense layer with 256 units, dropout of 0.25, and L2 regularization of 0.0001. The feature-combination search and the candidate-model comparison provide empirical support for the selected input structure and final model.

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ACKNOWLEDGMENT

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